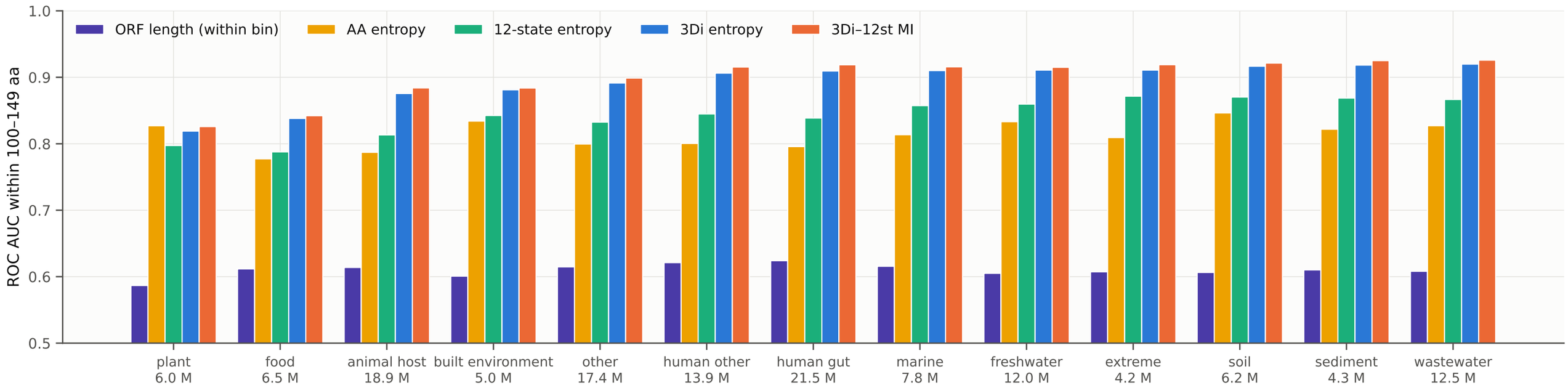
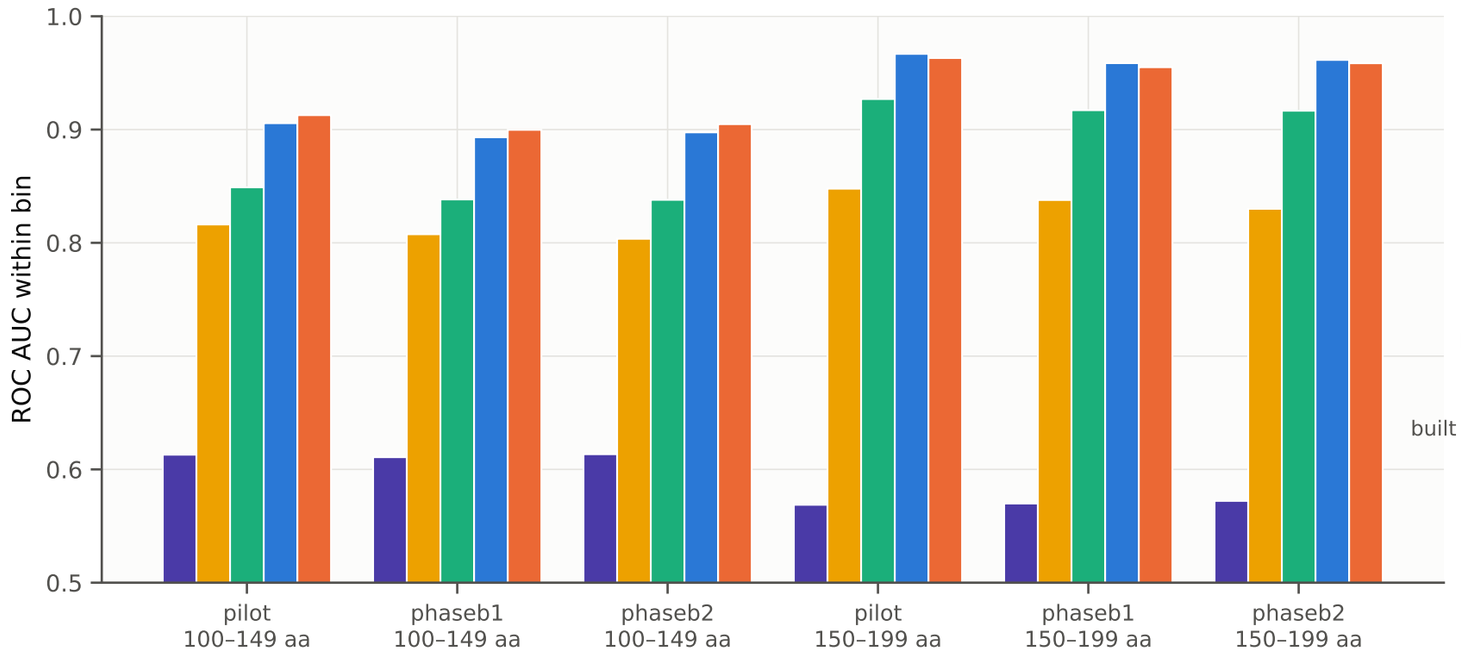
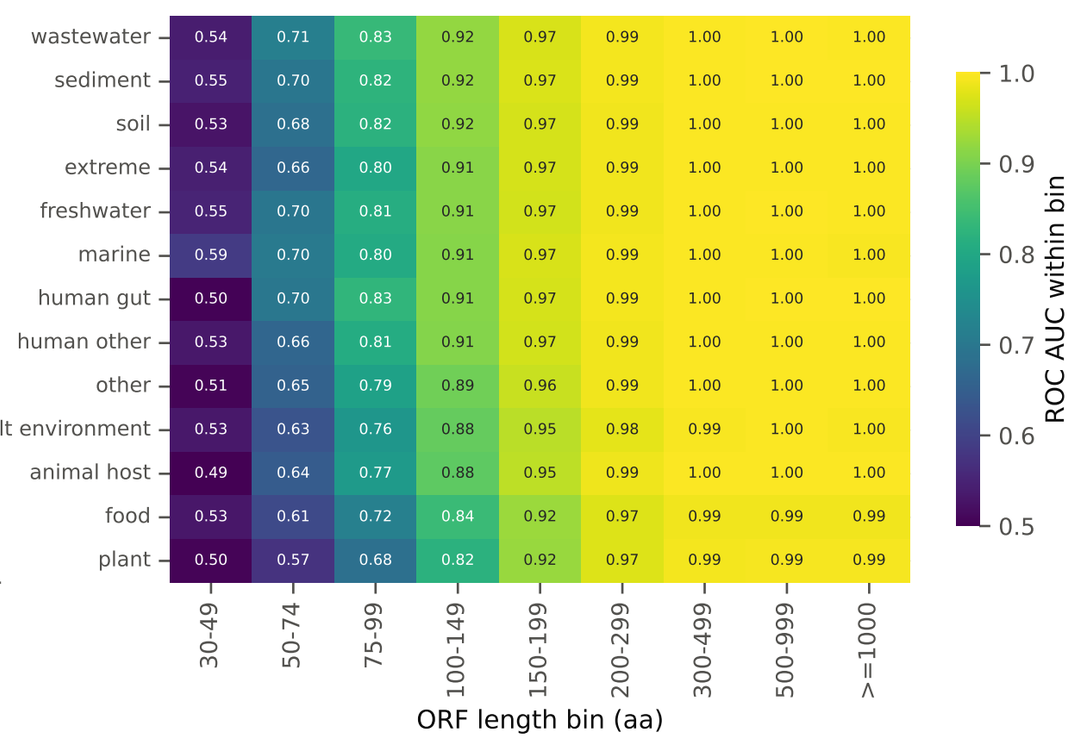
a


ORF length (within bin)
AA entropy
12-state entropy
3Di entropy
3Di–12st MI
ROC AUC within 100–149 aa
1.0
0.9
0.8
0.7
0.6
0.5
plant 6.0 M
food 6.5 M
animal host 18.9 M
built environment 5.0 M
other 17.4 M
human other 13.9 M
human gut 21.5 M
marine 7.8 M
freshwater 12.0 M
extreme 4.2 M
soil 6.2 M
sediment 4.3 M
wastewater 12.5 M

b


ROC AUC within bin
1.0
0.9
0.8
0.7
0.6
0.5
pilot 100–149 aa
phaseb1 100–149 aa
phaseb2 100–149 aa
pilot 150–199 aa
phaseb1 150–199 aa
phaseb2 150–199 aa

c


wastewater 0.54 0.71 0.83 0.92 0.97 0.99 1.00 1.00 1.00
sediment 0.55 0.70 0.82 0.92 0.97 0.99 1.00 1.00 1.00
soil 0.53 0.68 0.82 0.92 0.97 0.99 1.00 1.00 1.00
extreme 0.54 0.66 0.80 0.91 0.97 0.99 1.00 1.00 1.00
freshwater 0.55 0.70 0.81 0.91 0.97 0.99 1.00 1.00 1.00
marine 0.59 0.70 0.80 0.91 0.97 0.99 1.00 1.00 1.00
human gut 0.50 0.70 0.83 0.91 0.97 0.99 1.00 1.00 1.00
human other 0.53 0.66 0.81 0.91 0.97 0.99 1.00 1.00 1.00
other 0.51 0.65 0.79 0.89 0.96 0.99 1.00 1.00 1.00
built environment 0.53 0.63 0.76 0.88 0.95 0.98 0.99 1.00 1.00
animal host 0.49 0.64 0.77 0.88 0.95 0.99 1.00 1.00 1.00
food 0.53 0.61 0.72 0.84 0.92 0.97 0.99 0.99 0.99
plant 0.50 0.57 0.68 0.82 0.92 0.97 0.99 0.99 0.99
30-49
50-74
75-99
100-149
150-199
200-299
300-499
500-999
>=1000
ORF length bin (aa)
ROC AUC within bin
1.0
0.9
0.8
0.7
0.6
0.5